



EARTH AND SPACE SCIENCE 4

ACADEMIC

Grade 12

Curriculum Guide

EARTH AND SPACE SCIENCE 4

Pre-requisites: Earth and Space Science 1, Earth and Space Science 2, Earth and Space Science 3

Course Description:

Earth and Space Science 4 examines how natural hazards arise from geological and hydrometeorological processes and how these hazards can become disasters depending on the exposure, vulnerability, and capacity of communities. The learners study the components of disaster risk and use hazard maps, geospatial data, and historical records to assess disaster risks in local communities. They explore structural and nature-based mitigation strategies and analyze disaster risk reduction and management (DRRM) approaches implemented at community and national levels, including the provisions of Republic Act No. 10121. Through investigations and analyses of real-world data, the learners develop evidence-based recommendations to strengthen community preparedness and resilience.

The course also explores how energy systems, land-use practices, and infrastructure development influence environmental sustainability and disaster resilience in the Philippines. The learners analyze the Philippine energy mix and evaluate the role of renewable energy in supporting sustainable development. They investigate geologic risks associated with groundwater extraction and slope instability and assess geotechnical and land-management strategies that reduce landslide and subsidence risks. Using scientific data and geospatial tools, learners propose solutions that integrate sustainable energy systems, responsible land use, and disaster risk reduction strategies to support resilient and sustainable communities.

Elective: Academic

Time Allotment: 80 hours for one term

CONTENT	CONTENT STANDARDS <i>The learners demonstrate understanding that...</i>	LEARNING COMPETENCIES <i>The learners...</i>
Natural Hazards and Disaster Risk Reduction		
1. Natural hazards and disaster formation	1. natural hazards originate from geological and hydrometeorological processes and may become disasters depending on the exposure,	1. describe different types of geological and hydrometeorological hazards and their causes;
		2. explain how natural hazards become disasters by analyzing exposure, vulnerability, and capacity of communities;

CONTENT	CONTENT STANDARDS <i>The learners demonstrate understanding that...</i>	LEARNING COMPETENCIES <i>The learners...</i>
	vulnerability, and capacity of communities;	3. explain the components of disaster risk, including hazard, exposure, vulnerability, and capacity;
2. Hazard mapping and disaster risk assessment	2. hazard maps and geospatial information are important tools for identifying areas at risk and supporting disaster risk assessment;	4. use hazard maps and geospatial data to identify areas at risk in a local community; 5. conduct a basic disaster risk assessment for a selected community using available hazard maps and risk assessment tools;
3. Disaster risk reduction strategies	3. structural and nature-based mitigation strategies can reduce disaster risks and lessen the impacts of hazards on communities; and	6. compare structural and nature-based solutions used to mitigate disaster risks, including seismic retrofitting, flood barriers, dams, and mangrove restoration; 7. describe disaster risk reduction and management strategies used at the community, local government, and national levels;
4. Disaster governance and community resilience	4. disaster risk reduction policies and community preparedness plans help strengthen resilience and reduce disaster impacts.	8. use the provisions of Republic Act 10121 to explain how local DRRM councils implement disaster preparedness and response measures; 9. use hazard exposure data and historical disaster records to describe the effectiveness of existing community DRRM plans; and 10. create evidence-based strategies to improve disaster preparedness and risk reduction in a local community.
Performance Standards The learners can analyze hazard maps, disaster records, and risk data to assess disaster risks in local communities. They can evaluate disaster risk reduction strategies and policies and propose evidence-based recommendations that strengthen community preparedness, resilience, and disaster risk management.		
Suggested Performance Tasks Create a risk assessment report or presentation describing major hazards, risk levels, and possible mitigation strategies.		

CONTENT	CONTENT STANDARDS <i>The learners demonstrate understanding that...</i>	LEARNING COMPETENCIES <i>The learners...</i>
Develop a policy brief, hazard preparedness guide, or multimedia presentation that evaluates the effectiveness of existing community DRRM plans, recommends structural and nature-based mitigation strategies, and aligns proposed actions with the provisions of Republic Act No. 10121		
Energy Systems, Land Use, and Community Resilience		
5. Energy systems and sustainability	5. energy production and consumption influence environmental sustainability, and that the transition to renewable energy sources can help reduce environmental impacts and support sustainable development;	11. explain how energy production and consumption influence climate change and environmental sustainability in the Philippines;
		12. use information from Department of Energy (DOE) to describe the current Philippine energy mix and explain its environmental and socio-economic implications;
		13. use information from Department of Energy (DOE) and the International Renewable Energy Agency (IRENA) to evaluate the technical feasibility, environmental trade-offs, and socio-economic implications of solar, geothermal, and wind energy in the Philippine context;
		14. explain the strategies for increasing the share of renewable energy in the Philippine energy system;
6. Land subsidence and groundwater management	6. human activities such as excessive groundwater extraction can cause land subsidence and increase risks to communities and infrastructure;	15. explain how human land-use practices, including groundwater extraction and deforestation, influence geologic and environmental hazards;
		16. investigate ground subsidence associated with groundwater over-extraction using geospatial and monitoring data from PHIVOLCS and NAMRIA
7. Slope stability and geotechnical risk	7. geological conditions and human land-use practices influence slope stability, and appropriate geotechnical interventions can reduce landslide risks; and	17. use available geotechnical and hazard data to assess slope stability risks in a Philippine upland community
		18. explain appropriate geotechnical and land management interventions to reduce landslide and subsidence risks;

CONTENT	CONTENT STANDARDS <i>The learners demonstrate understanding that...</i>	LEARNING COMPETENCIES <i>The learners...</i>
8. Sustainable infrastructure and community resilience	8. sustainable energy systems, responsible land use, and resilient infrastructure contribute to disaster-resilient and sustainable communities.	19.evaluate how sustainable infrastructure, renewable energy systems, and land-use planning contribute to disaster-resilient communities; and
		20.create a community-level resilience plan integrating sustainable energy, responsible land management, and disaster risk reduction strategies.
Performance Standard <i>The learners can analyze energy, land-use, and geotechnical data to evaluate environmental risks and sustainability challenges affecting communities in the Philippines. They can assess renewable energy options, investigate land stability issues, and propose evidence-based strategies for developing sustainable and disaster-resilient communities.</i>		
Suggested Performance Tasks • Create an infographic, policy brief, or multimedia presentation that analyzes national energy data (e.g., from the Department of Energy) to describe the current Philippine energy mix. • Investigate the subsidence or landslide risks in a selected community using available hazard maps, geospatial data, or monitoring data.		